Section 1 Data Quality Checks

A1. How many bookings have invalid stays (checkout date ≤ check-in date)?

Ans. There are 120 invalid Stay Records in this dataset.

A2. Compute review rating range and mean SEPARATELY for each customer segment. What does this reveal? (1-line implication)

Customer Segment	Rating Min	Rating Max	Rating Mean
Corporate	1.5	5.0	3.62
Group	1.0	5.0	3.77
Individual	1.0	5.0	3.77

After normalizing the review scale, Group and Individual customers have slightly higher average ratings (3.77/5) than Corporate customers (3.62/5), indicating marginally lower satisfaction among Corporate guests

A3. After applying the relevant data-quality footnotes, what is realized revenue for property type = Luxury? (₹)

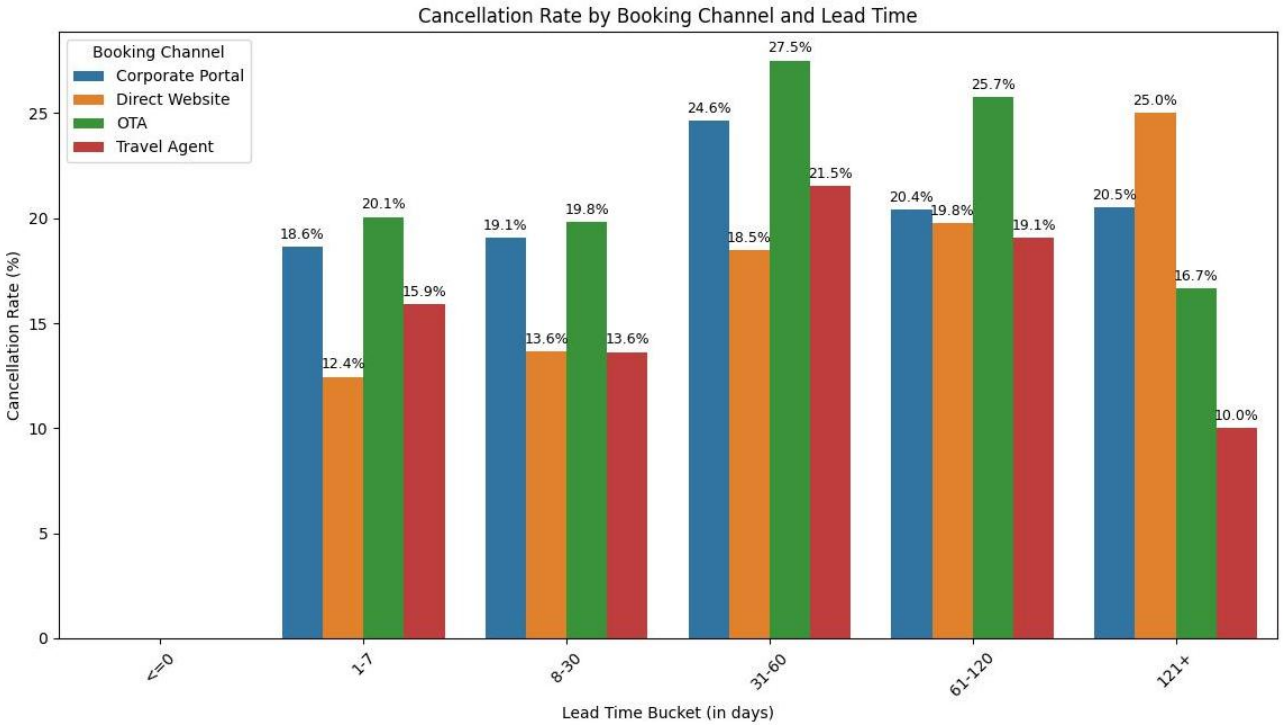
Total Revenue after applying Footnotes 1, 2, 3, and 8 is **₹89,009,636.79**

Section 2 Case Study: The Cancellation Crisis

Briefing: The Operations head has flagged that the platform's overall booking cancellation rate is around 19% — well above the industry benchmark of 12%. The board has set a target to reduce platform-wide cancellation by 5 percentage points (to ~14%) within two quarters. They expect you to identify where the problem is concentrated, diagnose the underlying cause from the data, and recommend a targeted intervention with a quantified expected impact. A vague "we should improve our policies" will not pass. Numbers and counterfactuals will.

A1. Cancellation Landscape

Chart 1: Cancellation Rate by Booking Channel and Lead Time.



OTA and Corporate Portal bookings have the highest cancellation rates **(22.1% and 20.2%),** while Direct Website bookings show the lowest cancellation rate **(14.9%)**, making them the most reliable booking channel.¶

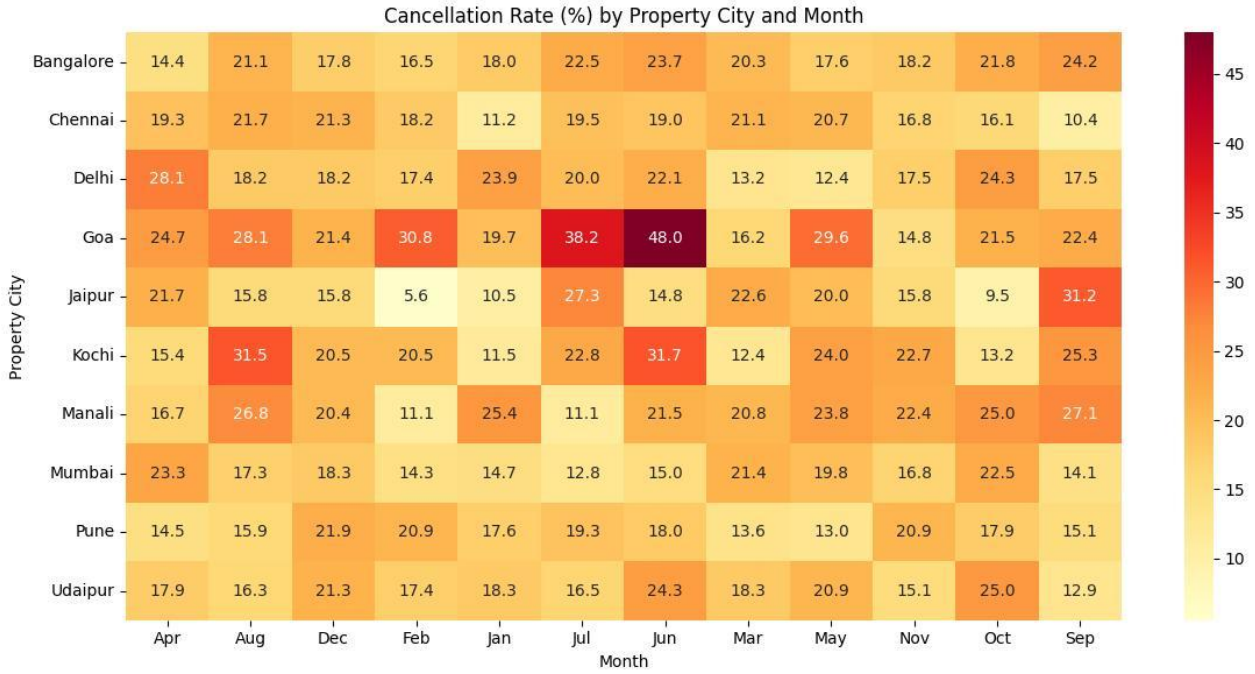
Platform cancellation rate: 19.18 %

Benchmark rate: 12.0 %

Target rate: 14.0 %

Gap to benchmark: 7.18 percentage points

Chart 2 : Cancellation rate(%) by Property City and Month



This is a strong finding — **Goa in June/July/August shows 37–53%** cancellation vs. a 19% platform baseline, and the city - month combined slice spreads 52.6pp while no single dimension alone exceeds 9pp. That's a real interaction effect, not just a city problem or a seasonality problem in isolation.

The dimension that drives the most variance in cancellation rate is **Property City × Calendar Month**.

Cancellation rates range from 0.00% (Jaipur in March) to 52.63% (Goa in June), producing a spread of **52.63 percentage points**.

This spread is substantially larger than the variation observed across booking channels (7.22 percentage points), customer segments (0.82 percentage points), and loyalty tiers (7.45 percentage points).

This indicates that cancellation behavior is concentrated in specific locations and time periods rather than being uniformly distributed across customer or booking characteristics.

The slice that most strongly separates high-cancel from low-cancel bookings is property city by calendar month, which shows a much wider cancellation spread than booking channel by lead-time bucket.

A2 Rate vs. Volume.

1. Top 3 Slices by Cancellation Rate(worst-performing)

Property City	Booking Month	Total Bookings	Cancelled Bookings	Cancellation Rate (%)
Goa	Jun	75	36	48.00
Goa	Jul	76	29	38.16
Kochi	Jun	60	19	31.67

2. Top 3 Slices by Cancellation Count

Property City	Booking Month	Cancelled Bookings	Total Bookings	Cancellation Rate (%)
Pune	Nov	37	177	20.90
Chennai	Mar	36	171	21.05
Goa	Jun	36	75	48.00

High-rate slices and high-volume slices are often not the same. A slice can have a very high cancellation rate but relatively few bookings, resulting in a small number of cancellations overall.

This distinction matters because the board's goal is to reduce the platform-wide cancellation rate by 5 percentage points. To achieve that, interventions should focus on slices with the largest number of cancellations (high volume), since reducing cancellations in small-volume, high-rate segments may have little impact on the overall platform metric.¶

A3 Root Cause: Hypothesis Testing.

Hypothesis 1: High Cancellation in Goa (June) is Driven by Lead Time

Question:

Are customers booking much further in advance in Goa during June, leading to more cancellations?

Metric	Goa June	Rest of Platform
Mean Lead Time	38.21 days	32.52 days
Median Lead Time	15.00 days	15.00 days

T-statistic: 0.404

P-value: 0.6875

Since:

p-value = 0.6875 > 0.05

We fail to reject the null hypothesis.

The difference in lead time is not statistically significant.

Although Goa-June bookings have a slightly higher average lead time (38 vs 33 days), the distributions are very similar and the difference is likely due to random variation.

Hypothesis rejected.

Executive Summary:

Goa–June bookings had a higher average lead time (38.2 days) than the rest of the platform (32.5 days), though both groups had the same median lead time (15 days). However, a Welch's t-test returned a p-value of 0.688, indicating no statistically significant difference. Therefore, lead time does not appear to be the primary driver of Goa's unusually high June cancellation rate.

Hypothesis 2: Channel-Mix Effect

Question:

Is Goa's high cancellation rate simply because it receives a larger share of bookings from channels that typically have high cancellation rates?

Channel Distribution Comparison

Booking Channel	Goa June (%)	Rest of Platform (%)
OTA	36.84	35.37
Corporate Portal	28.07	26.77
Travel Agent	22.81	20.10
Direct Website	12.28	17.76

Platform-Wide Cancellation Rate by Channel

Booking Channel	Bookings	Cancellations	Cancellation Rate (%)
OTA	4,245	937	22.07
Corporate Portal	3,213	650	20.23
Travel Agent	2,414	399	16.53
Direct Website	2,128	316	14.85

Interpretation:

If channel mix were the primary driver, we would expect Goa June to be heavily concentrated in high-cancellation channels compared with the rest of the platform.

More importantly:

The highest channel cancellation rate on the platform is only 22.07% (OTA).

Goa June's cancellation rate is 52.63%.

Even if Goa June had a slightly higher concentration of OTA bookings, that cannot explain why its cancellation rate is more than double the highest channel-level cancellation rate.

Quantifying the Effect:

Expected cancellation rate using Goa June's channel mix:

$(36.84\% \times 22.07\%)+(28.07\% \times 20.23\%)+(22.81\% \times 16.53\%)+(12.28\% \times 14.85\%)$

$=17.74\%$

Platform average: 19.04%

Observed Goa June: 52.63%

Comparison:

Metric	**Cancellation Rate**
Platform Average	19.04%
Expected from Goa Channel Mix	17.74%
Actual Goa June	52.63%

The channel composition would predict roughly 18% cancellations, not 53%.

Hypothesis 2 Rejected:

Although Goa June contains slightly fewer Direct Website bookings and slightly more OTA/Corporate bookings than the rest of the platform, the differences are too small to explain the extreme cancellation rate. The channel mix predicts a cancellation rate near 18%, whereas the observed rate is 52.63%. **Therefore, booking-channel composition is not the root cause of the Goa-June cancellation hotspot.**

Hypothesis 3: Genuine City + Season Effect

Question:

Is June uniquely problematic in Goa, beyond what we observe:

Goa Cancellation Rate by Month

Month	Bookings	Cancellations	Cancellation Rate (%)
June	57	30	52.63
August	88	39	44.32
July	65	24	36.92
March	70	21	30.00
September	81	19	23.46
February	77	18	23.38
October	76	17	22.37
November	72	15	20.83
May	80	16	20.00
December	75	13	17.33
January	73	11	15.07
April	77	11	14.29

Goa June: 52.63

Goa Other Months: 24.46

June Cancellation Rate by Property City

Property City	Bookings	Cancellations	Cancellation Rate (%)
Goa	57	30	52.63
Kochi	70	22	31.43
Manali	67	17	25.37
Jaipur	33	7	21.21

Property City	Bookings	Cancellations	Cancellation Rate (%)
Chennai	160	33	20.62
Udaipur	129	26	20.16
Bangalore	137	25	18.25
Mumbai	107	19	17.76
Delhi	80	12	15.00
Pune	170	25	14.71

p-value: 0.0

For Hypothesis 3 (Genuine City + Season Effect), a p-value of 0.0 (effectively $p < 0.0001$) is extremely strong evidence against the null hypothesis.

Hypothesis Test:

1. H_0 (Null Hypothesis):

The cancellation rate in Goa during June is the same as the cancellation rate in the rest of the platform.

2. H_1 (Alternative Hypothesis):

The cancellation rate in Goa during June is significantly different from the rest of the platform.

Result:

p-value < 0.0001

Since: p-value << 0.05

we reject the null hypothesis.

Interpretation:

The extremely small p-value indicates that the difference between:

Goa in June: 52.63% cancellation rate

Rest of Platform: 19.04% cancellation rate

is not due to random chance.

The probability of observing such a large difference if there were actually no real Goa-June effect is effectively zero.

Comparison between hypothesis:

Hypothesis	Evidence	Verdict
Lead-Time Effect	Goa–June mean lead time (38.2 days) was not significantly different from the rest of the platform (Welch's t-test, p = 0.6875).	Rejected
Channel-Mix Effect	Goa's booking-channel composition would predict a cancellation rate of only **17.74%**, far below the observed **52.63%**.	Rejected
City + Season Effect	Goa–June recorded a **52.63%** cancellation rate, significantly above the platform average; χ^2 test found a highly significant association ($p < 0.0001$).	Strongly Supported

Final Root Cause Statement

The data most strongly support a city-and-season interaction effect. Goa experiences an exceptionally high cancellation rate during June (52.63%), far exceeding the platform average of 19.04%. Alternative explanations were tested and ruled out: lead times in Goa June were not significantly different from the rest of the platform (p = 0.6875), and the booking-channel composition was broadly similar, predicting only a 17.74% cancellation rate. In contrast, a chi-square test comparing Goa June with the rest of the platform produced a p-value below 0.0001, confirming that the elevated cancellation rate is statistically significant and reflects a genuine Goa-specific seasonal phenomenon rather than booking behavior or channel mix.

Executive Recommendation:

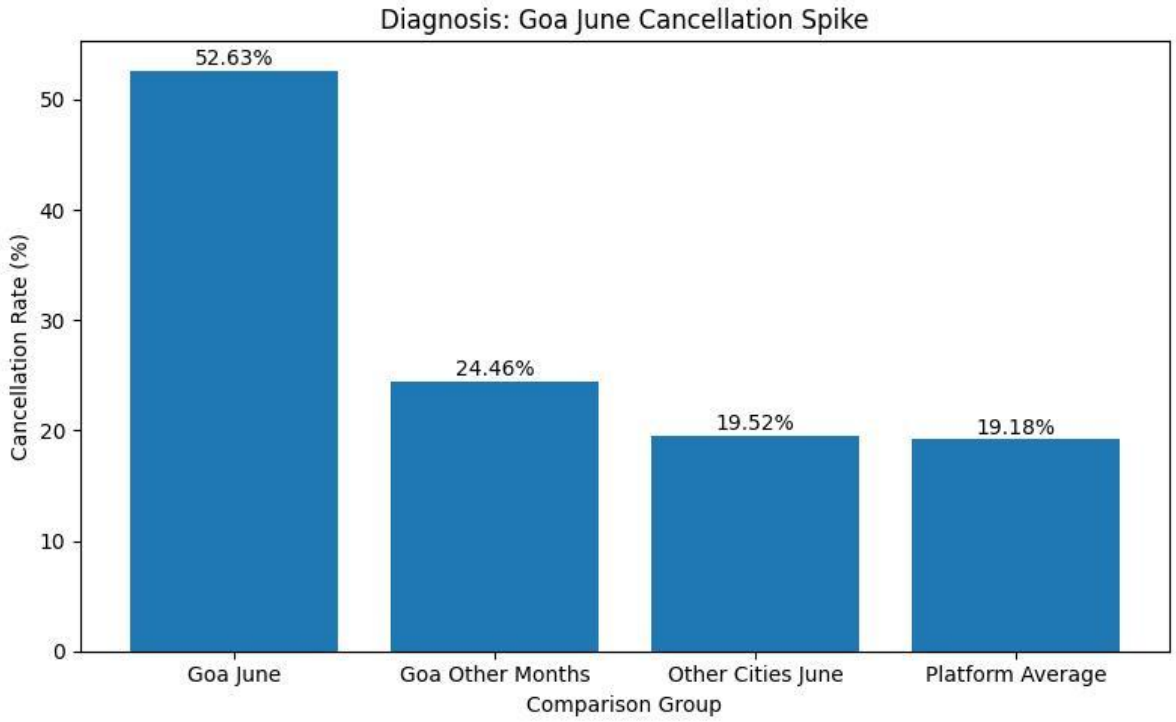
Since the **root cause appears to be Goa during June specifically**

Interventions should be targeted at this slice rather than platform-wide:

This is the strongest evidence-based conclusion from the hypothesis testing section of the case study.

A4. Targeted Recommendation

Recommendation:



Introduce a tiered cancellation policy with a 20% non-refundable deposit for bookings in Goa during June.

Target Segment (Who)

- Property City: Goa
- Stay Month: June
- All booking channels

This segment is targeted because it represents the most severe cancellation hotspot on the platform, with a cancellation rate of 52.63%, nearly three times the platform average of 19.04%.

Proposed Policy (What)

Require customers booking stays in Goa during June to pay a 20% non-refundable deposit at the time of booking.

Customers would retain the flexibility to cancel, but the deposit would not be refunded.

This policy is designed to:

- Reduce speculative bookings.
- Increase customer commitment.
- Discourage low-intent reservations.
- Recover a portion of revenue from unavoidable cancellations.

Quantified Expected Impact

Current Goa-June bookings:

- Total bookings = 57
- Cancelled bookings = 30
- Cancellation rate = 52.63%

Assuming the deposit policy reduces cancellations within this segment by 20%–40% (a conservative range commonly observed when financial commitment is introduced):

- Prevented cancellations = 30 × 20% to 40%
- Prevented cancellations = 6 to 12 bookings

Platform-wide impact:

If total platform bookings = 12,000,

Reduction in cancellation rate:

(6 ÷ 12,000) × 100 = 0.05 pp

to

(12 ÷ 12,000) × 100 = 0.10 pp

Expected platform-wide reduction:
0.05–0.10 percentage points

While the direct impact is modest because Goa-June represents a relatively small share of platform volume, this intervention targets the single highest-risk slice identified in the analysis and provides a scalable pilot that can later be extended to other high-cancellation city-month combinations.

Risk / Collateral Effect:

A stricter cancellation policy may reduce booking conversion rates among price-sensitive travelers who prefer flexible booking options. Some demand may shift to competitors offering free cancellation, potentially reducing booking volume within the targeted segment.

Section 3 SQL Challenge

Schema Design Task :

Design a normalized schema by writing CREATE TABLE statements for at least three tables (customers, properties, bookings, optionally reviews).

Code –

```
CREATE DATABASE Hotel_booking;

USE Hotel_booking;


-- CUSTOMERS TABLE

CREATE TABLE customers (

        customer_id INT PRIMARY KEY,

  customer_name VARCHAR(100) NOT NULL,

  customer_segment VARCHAR(20) NOT NULL,

  customer_signup_date DATE NOT NULL,

  customer_home_city VARCHAR(50) NOT NULL,

  customer_loyalty_tier VARCHAR(20) NOT NULL,


  CHECK (customer_segment IN ('Individual','Group','Corporate')),

  CHECK (customer_loyalty_tier IN ('None','Silver','Gold','Platinum'))

);


-- PROPERTIES TABLE

CREATE TABLE properties (

   property_id INT PRIMARY KEY,

  property_name VARCHAR(100) NOT NULL,

  property_city VARCHAR(50) NOT NULL,

  property_star_rating DECIMAL(2,1) NOT NULL,

  property_type VARCHAR(20) NOT NULL,

  property_total_rooms INT NOT NULL,


  CHECK (property_star_rating BETWEEN 1 AND 5),

  CHECK (property_total_rooms > 0),
```

```
  UNIQUE (property_name, property_city)
);

-- BOOKINGS TABLE
CREATE TABLE bookings (
  booking_id INT PRIMARY KEY,

  customer_id INT NOT NULL,
  property_id INT NOT NULL,

  booking_date DATE NOT NULL,
  checkin_date DATE NOT NULL,
  checkout_date DATE NOT NULL,

  room_type VARCHAR(20) NOT NULL,
  num_rooms INT NOT NULL,
  nights INT NOT NULL,

  booking_channel VARCHAR(30) NOT NULL,

  adr DECIMAL(10,2) NOT NULL,
  discount_amount DECIMAL(10,2) DEFAULT 0,

  coupon_code VARCHAR(30),

  total_amount DECIMAL(12,2) NOT NULL,

  payment_method VARCHAR(30) NOT NULL,

  booking_status VARCHAR(20) NOT NULL,

  FOREIGN KEY (customer_id)
    REFERENCES customers(customer_id),
```

```
  FOREIGN KEY (property_id)
    REFERENCES properties(property_id),

  CHECK (checkout_date > checkin_date),
  CHECK (booking_date >= '2000-01-01'),
  CHECK (num_rooms > 0),
  CHECK (nights > 0),
  CHECK (total_amount >= 0),
  CHECK (discount_amount >= 0),

  CHECK (
    booking_status IN
    ('Completed','Cancelled','No-Show')
  )
);
```

A-Q1.

For each property city, return the property id, property name, and revenue of the property with the HIGHEST realized revenue. Use RANK () OVER (PARTITION BY property city ORDER BY revenue DESC). Group correctly given Footnote.

```
WITH property_revenue AS (
  SELECT
    p.property_city,
    p.property_id,
    p.property_name,
    SUM(b.total_amount) AS revenue
  FROM bookings b
  JOIN properties p
    ON b.property_id = p.property_id
  WHERE b.booking_status = 'Completed'
  GROUP BY
    p.property_city,
    p.property_id,
    p.property_name
),
```

```
ranked_properties AS (
  SELECT
    property_city,
    property_id,
    property_name,
    revenue,
    RANK() OVER (
      PARTITION BY property_city
      ORDER BY revenue DESC
    ) AS rnk
  FROM property_revenue
)

SELECT
  property_city,
  property_id,
  property_name,
  ROUND(revenue,2) AS revenue
FROM ranked_properties
WHERE rnk = 1
ORDER BY property_city;
```

Output -

Property City	Property ID	Property Name	Revenue (₹)
Bangalore	2	The Grand Plaza	13,957,983.73
Chennai	53	Saffron Heights	14,355,166.46
Delhi	49	Mango Retreat	4,200,186.89
Goa	23	Sapphire Retreat	19,619,575.91
Jaipur	30	Ivory Retreat	1,805,426.61
Kochi	1	The Grand Plaza	6,631,787.86
Manali	10	Lotus Manor	4,347,636.54

Property City	Property ID	Property Name	Revenue (₹)
Mumbai	11	Coral Palace	10,719,830.16
Pune	34	Velvet Palace	6,062,114.47
Udaipur	6	Hilltop Inn	15,669,688.12

A-Q2.

Using LAG () over each customer's Completed bookings ordered by check-in date, compute the gap in days between consecutive bookings per customer. Return the COUNT of customers whose AVERAGE gap is under 30 days.

```
WITH completed_bookings AS (
  SELECT
    customer_id,
    checkin_date,
    LAG(checkin_date) OVER (
      PARTITION BY customer_id
      ORDER BY checkin_date
    ) AS prev_checkin
  FROM bookings
  WHERE booking_status = 'Completed'
),

booking_gaps AS (
  SELECT
    customer_id,
    DATEDIFF(checkin_date, prev_checkin) AS gap_days
  FROM completed_bookings
  WHERE prev_checkin IS NOT NULL
),

avg_gap_per_customer AS (
  SELECT
    customer_id,
    AVG(gap_days) AS avg_gap_days
  FROM booking_gaps
```

```
  GROUP BY customer_id
)

SELECT COUNT(*) AS customers_avg_gap_under_30_days
FROM avg_gap_per_customer
WHERE avg_gap_days < 30;
```

customers_avg_gap_under_30_days
0